

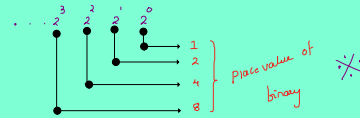
Day-1

Number system		Base value	Place value
Binary	0-1	2	* - CNA
Decimal	0-9	10	
Hexa	0-15	16	
	1		
	2		
	3		
	4		
	5		
	6		
	7		
	8		
	9		
	A		
	B		
	C		
	D		
	E		
	F		

Introduction
Network

Finding place value of binary system

Base value = 2



Conversion

Decimal $\longleftrightarrow$ Binary

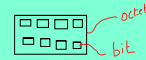
Hexa $\longleftrightarrow$ Binary

Decimal to Binary

place value | octet = 8 bit

2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
128	64	32	16	8	4	2	1

9	→	0 0 0 0 1 0 0 1
20	→	0 0 0 1 0 1 0 0
128	→	1 0 0 0 0 0 0 0
192	→	1 1 0 0 0 0 0 0
224	→	1 1 1 0 0 0 0 0
240	→	1 1 1 1 0 0 0 0
248	→	1 1 1 1 1 0 0 0
254	→	1 1 1 1 1 1 0 0
255	→	1 1 1 1 1 1 1 1



Hexa $\rightarrow$ Binary

1 hexa = 4 bit

8	4	2	1
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A	→	1 0 1 0
B	→	
C	→	
D	→	
E	→	
F	→	1 1 1 1

10 IP

IP Decimal

MAC Hexa

2 A

00001010



Binary

0010 1010